Integral bases, Galois groups, and class groups of the trinomials $x^n - ax - a$

Abstract

For integers $n \geq 2$ and $a \neq 0$, write $f_{n,a} = x^n - ax - a$. Whenever $f_{n,a}$ is irreducible, let θ be a root and $K_{n,a} = \mathbb{Q}(\theta)$. The relation $\theta^n = a(\theta + 1)$ determines the arithmetic of $K_{n,a}$ almost completely. We compute the discriminant, the factorization of every rational prime, and the exact power of every prime dividing the index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$, excluding one irregular local case. As consequences, the Galois group is determined for every irreducible member of degree 3, 4, and 5; the element $\theta + 1$ is shown to be a unit of norm one, and to be primitive in the unit group for every squarefree cubic parameter with $2 \leq |a| \leq 33535$; and the class number is shown to be divisible by an explicit genus number built from the prime factors of a. The cyclic fields of degree three are identified with Shanks' simplest cubic fields. The local statements were checked on 6388 fields and 25287 ramified primes, and the class-group statements on 1878 fields.

1 Introduction

Let $n \geq 2$ and $a \in \mathbb{Z} \setminus \{0\}$, and set

$$f_{n,a}(x) = x^n - ax - a, \qquad D_n(a) = n^n + (-1)^n (n-1)^{n-1} a.$$

If $f_{n,a}$ is irreducible, let θ be a root and write $K_{n,a} = \mathbb{Q}(\theta)$. The polynomial is rigid: its two non-leading coefficients are equal, so a root satisfies $\theta^n = a(\theta + 1)$. Every prime divisor of a therefore ramifies in a way that can be read off from the valuation $v_p(a)$, and $\theta + 1$ is a canonical unit.

Trinomials have been studied from several directions. Jakhar, Khanduja, and Sangwan gave coefficient criteria for the ring $\mathbb{Z}[\theta]$ to be the full ring of integers of a general trinomial extension; Jones and White constructed infinite monogenic families $x^n + Ax^m + A$ with non-squarefree discriminant [6]; and Ibarra, Lembeck, Ozaslan, Smith, and Stange treated monogenity for $x^n + ax + b$ in small degree [5]. The Galois group of a prime-degree trinomial $X^p + aX^s + a$ was determined in almost all cases by Bensebaa, Movahhedi, and Salinier [1]. Genus fields of non-Galois extensions go back to Ishida [3, 4], and the statistics of the resulting class-number divisibility have been taken up by McGown, Thorne, and Tucker [7]. What is missing is a closed-form description, uniform in the degree, of the single family in which the middle coefficient equals the constant term.

The paper is organized as follows. Section 2 gives the discriminant and the splitting of every rational prime. Section 3 computes the index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$. Section 4 determines the Galois group in degrees 3, 4, and 5. Section 5 treats the unit $\theta + 1$, and Section 6 the genus contribution to the class group. Section 7 identifies the cyclic cubic members with Shanks' fields. Section 8 records the computations.

2 Discriminant and splitting

Proposition 1. $\operatorname{disc}(f_{n,a}) = (-1)^{(n-1)(n-2)/2}\, a^{n-1} D_n(a)$.

Proof. The discriminant of $x^n + Ax + B$ is

$$(-1)^{n(n-1)/2}\Big(n^n B^{n-1} + (-1)^{n-1}(n-1)^{n-1} A^n\Big).$$

Substituting $A = B = -a$ and simplifying the sign $(-1)^{n(n-1)/2}(-1)^{n-1} = (-1)^{(n-1)(n-2)/2}$ gives the formula. Both expressions were compared for $2 \le n \le 30$ and $1 \le |a| \le 5$. □

In particular $f_{n,a}$ and $f'_{n,a}$ share a root modulo p if and only if p divides $aD_n(a)$. Indeed $nc^{n-1} = a$ and $c^n = a(c+1)$ eliminate to $D_n(a) \equiv 0$, unless p already divides a.

The splitting law is a Newton-polygon computation in the sense of Ore [8]. We use it in the following form. Let $f \in \mathbb{Z}[x]$ be monic and irreducible, and let p be a prime whose Newton polygon consists of one side S of slope $-h/e$ in lowest terms, with residual polynomial $\overline{R} \in \mathbb{F}_p[Y]$ separable of degree d. If $\overline{R} = \prod_j \overline{R}_j$ is the factorization into irreducibles, then

$$p\mathcal{O}_K = \prod_j \mathfrak{p}_j^{e}, \qquad f(\mathfrak{p}_j/p) = \deg \overline{R}_j.$$

For a side joining $(0,k)$ to $(n,0)$ on which no other point of the coefficient sequence lies, one has $e = n/\gcd(k,n)$ and $\overline{R}(Y) = Y^{\gcd(k,n)} - \overline{u}$, where u is the prime-to-p part of the constant term.

Theorem 2. *Let $f_{n,a}$ be irreducible, let $K = K_{n,a}$, and let p be a prime.*

(i) *Suppose $p \mid a$. Write $k = v_p(a)$, $g = \gcd(k,n)$, and $u = ap^{-k} \bmod p$, and assume $p \nmid g$. Then $p\mathcal{O}_K = \mathfrak{p}_1^{n/g} \cdots \mathfrak{p}_r^{n/g}$, where the residue degrees $f(\mathfrak{p}_i/p)$ are the degrees of the irreducible factors of $Y^g - u$ over $\mathbb{F}_p$.*

(ii) *Suppose $p \nmid aD_n(a)$. Then p splits in K according to the factorization of $f_{n,a}$ modulo p.*

(iii) *Suppose $p \nmid a$ and $k = v_p\big(D_n(a)\big) \ge 1$. Then $p \nmid n(n-1)$. The reduction of $f_{n,a}$ has a double root $c \equiv -n(n-1)^{-1} \pmod p$ and no other repeated root, and*

$$p\mathcal{O}_K = \mathfrak{r}\,\mathfrak{p}_1 \cdots \mathfrak{p}_s,$$

where the $\mathfrak{p}_i$ are prime ideals of ramification index 1 whose residue degrees are the degrees of the irreducible factors of $f_{n,a}(x)/(x-c)^2$ over $\mathbb{F}_p$, and where the factor $\mathfrak{r}$ is

- *a prime of ramification index 2 and residue degree 1, if k is odd;*
- *a product of two distinct primes of residue degree 1, if k is even and $\Big(\dfrac{-2(n-1)\,n^{n-1} D_n(a)\,p^{-k}}{p}\Big) = 1$;*
- *a prime of residue degree 2, if k is even and the same symbol equals -1.*

Proof. (i). The coefficients of $f_{n,a}$ have valuations 0 (at x^n), k (at x), and k (at x^0). The point $(1,k)$ lies strictly above the segment from $(0,k)$ to $(n,0)$, because the height of that segment at the first abscissa is $k(n-1)/n < k$. Ore's theorem applies as soon as the residual polynomial $Y^g - u$ is separable. Its derivative is gY^{g-1}, so a common root would force $p \mid g$ or $u = 0$. Both are excluded: $u \not\equiv 0$ by construction of k, and $p \nmid g$ by hypothesis. The stated factorization follows.

(ii). Such a prime divides neither the index, by Theorem 3 below, nor the discriminant, so Dedekind's criterion reduces the splitting to the factorization of $f_{n,a}$ modulo p.

(iii). A common root of $f_{n,a}$ and $f'_{n,a} = nx^{n-1} - a$ modulo p satisfies $nc^{n-1} \equiv a$ and $c^n \equiv a(c+1)$. Eliminating a gives $D_n(a) \equiv 0$, and solving gives $c \equiv -n(n-1)^{-1}$. In particular $p \nmid n(n-1)$, and $c \not\equiv 0$. The second derivative $f''(c) = n(n-1)c^{n-2}$ is nonzero modulo p, so the root is exactly double and the cofactor is separable modulo p.

Translate $x = c + Y$ with $c \in \mathbb{Z}$ a lift of this root for which $v_p\big(f(c)\big) = k$ and $v_p\big(f'(c)\big) \geq \lceil k/2 \rceil$; such a lift exists because $D_n(a)$ is, up to a p-unit, the resultant of f and f', hence the precise valuation of the double root. The Newton polygon of the translated polynomial then has a single side of horizontal length 2 and slope $-k/2$, the remaining roots contributing sides of slope 0. The residual polynomial of the steep side is the quadratic

$$T^2 - \gamma, \qquad \gamma \equiv -2(n-1)\, n^{n-1} D_n(a)\, p^{-k} \pmod{p},$$

up to squares in $\mathbb{F}_p^\times$: the leading contribution is $\frac{1}{2}f''(c) = \frac{1}{2}n(n-1)c^{n-2}$, and substituting $c \equiv -n(n-1)^{-1}$ together with $a \equiv n^n(n-1)^{1-n}(-1)^n$ produces the displayed class. If k is odd the slope is $-k/2$ in lowest terms with denominator 2, so the side contributes a unique prime of ramification index 2 and residue degree 1. If k is even the slope is an integer, the ramification index is 1, and the residual quadratic splits, is irreducible, or has a double root according to the Legendre symbol of γ. It has no double root because that symbol is ± 1. □

3 The index

Theorem 3. *Let $f_{n,a}$ be irreducible and let $K = K_{n,a}$. For a prime p,*

$$v_p\big([\mathcal{O}_K : \mathbb{Z}[\theta]]\big) = \begin{cases} \dfrac{(k-1)(n-1)+g-1}{2}, & \text{if } p \mid a,\ k = v_p(a),\ g = \gcd(k,n),\ p \nmid g, \\ \lfloor k/2 \rfloor, & \text{if } p \nmid a,\ k = v_p\big(D_n(a)\big). \end{cases}$$

In particular $\mathbb{Z}[\theta] = \mathcal{O}_K$ whenever a is squarefree, no prime divisor of a divides n, and $D_n(a)$ is squarefree.

Proof. Ore's formula expresses $v_p\big([\mathcal{O}_K : \mathbb{Z}[\theta]]\big)$ as a sum of local contributions of the sides of the Newton polygon. On a side of horizontal length ℓ, denominator e, and residual polynomial a product of r distinct irreducible factors, the contribution is $\frac{1}{2}\big(\ell(v-1) + e\cdot(\text{number of residual factors}) - \ell/e\big)$ in the separable case, which simplifies to the two displayed quantities. In case $p \mid a$ with $p \nmid g$ there is one side, of length n and denominator n/g, with g residual degree in total, and the formula returns $\big((k-1)(n-1)+g-1\big)/2$. In case $p \nmid a$ the only contribution is the side of length 2 and height k from Theorem 2(iii), which contributes $\lfloor k/2 \rfloor$. Every other prime contributes nothing. □

The hypothesis $p \nmid g$ is essential. If a prime divides both n and $v_p(a)$, the residual polynomial is inseparable and the index depends on higher digits of a. For a concrete illustration, $f_{6,-320}$ is irreducible and the power of 2 in its index is 4, which neither row of the theorem predicts.

4 Galois groups in degrees 3, 4, and 5

Theorem 4. *The polynomial $f_{3,a}$ is reducible if and only if $a = 8$, in which case $f_{3,8} = (x+2)(x^2-2x-4)$. Otherwise $\mathrm{Gal}(f_{3,a}) \cong C_3$ if $4a - 27$ is a square, and $\mathrm{Gal}(f_{3,a}) \cong S_3$ if it is not. The square case occurs precisely for $a = t^2 + 3t + 9$ with $t \in \mathbb{Z}$.*

Proof. A rational root r is an integer divisor of a with $r^3 = a(r+1)$, so $r+1$ divides $r^3 = \big((r+1)-1\big)^3$ and hence divides 1. Thus $r \in \{0, -2\}$. The value $r = 0$ forces $a = 0$, and $r = -2$ forces $a = 8$. Degree three admits no other factorization type. For $a \neq 8$, Proposition 1 gives $\operatorname{disc}(f_{3,a}) = a^2(4a - 27)$, which is a square if and only if $4a - 27$ is a square. An irreducible cubic has Galois group C_3 or S_3 according as its discriminant is or is not a square.

If $4a - 27 = m^2$, then m is odd, because m even would make $m^2 + 27 \equiv 3 \pmod 4$. Writing $m = 2t + 3$ yields $a = t^2 + 3t + 9$. Conversely this value of a gives $4a - 27 = (2t + 3)^2$. None of these equals 8. □

The same fields are identified with Shanks' family in Section 7.

Theorem 5. *Let $a \neq 0$ and let $R_a(y) = y^3 + 4ay - a^2$ be the cubic resolvent of $f_{4,a}$.*

(i) *$f_{4,a}$ is reducible if and only if $a = -16$, in which case $f_{4,-16} = (x + 2)(x^3 - 2x^2 + 4x + 8)$.*

(ii) *For every $a \neq 0$, $\operatorname{disc}(f_{4,a})$ is not a square. Consequently the Galois group is never A_4 or V_4.*

(iii) *R_a has a rational root if and only if*

$$a = (s - 2)(s + 2)^2 \qquad \text{or} \qquad a = -(s - 2)^2(s + 2)$$

for some integer $s \geq 0$ with $s \neq 2$. In that case $f_{4,a}$ is irreducible except for $a = -16$, the group is C_4 for $a = -5$, and the group is D_4 for every other such a.

(iv) *If R_a is irreducible, then $\operatorname{Gal}(f_{4,a}) \cong S_4$.*

Proof. The resolvent of $x^4 + px^2 + qx + r$ is $y^3 - py^2 - 4ry + (4pr - q^2)$. For $f_{4,a}$ one has $p = 0$, $q = -a$, $r = -a$, which gives R_a.

(i). A linear factor produces a rational root r with $r^4 = a(r + 1)$, so again $r + 1$ divides 1 and $r \in \{0, -2\}$. Only $r = -2$ occurs, giving $a = -16$, and the factorization is the one stated. Suppose instead $f_{4,a} = (x^2 + px + q)(x^2 - px + s)$ with $p, q, s \in \mathbb{Z}$. Comparing coefficients gives $q + s = p^2$, $p(s - q) = -a$, and $qs = -a$. If $p = 0$ then $a = 0$. If $p \neq 0$, then $a = p^2\big(2 \pm \sqrt{p^2 + 4}\big)$, so $p^2 + 4$ is a square, say w^2. Then $w^2 - p^2 = 4$. The integer factorizations of 4 give only the solution $p = 0$, which was excluded. Thus there is no quadratic-quadratic factorization.

(ii). By Proposition 1, $\operatorname{disc}(f_{4,a}) = -a^3(27a + 256)$. A negative integer is not a square, so it is enough to rule out $a < 0$ with $27a + 256 > 0$, that is $a \in \{-1, \ldots, -9\}$, together with the observation that $a \leq -10$ makes $27a + 256 < 0$ and $a > 0$ makes the discriminant negative. The nine values $-a^3(27a + 256)$ are 229, 1616, 4725, 9472, 15125, 20304, 22981, 20480, and 9477, none of which is a square. Groups A_4 and V_4 have square discriminant, so neither occurs. An irreducible quartic whose resolvent is irreducible therefore has group S_4, which is (iv).

(iii). Suppose $R_a(r) = 0$ for an integer r. The quadratic formula in a gives $a = 2r \pm \sqrt{r^2(r + 4)}$, so $r^2(r+4)$ must be a square. Since r^2 is already a square, $r+4$ itself is a square: $r = s^2 - 4$ for some integer $s \geq 0$. Substituting and simplifying produces exactly the two families $a = (s - 2)(s + 2)^2$ and $a = -(s - 2)^2(s + 2)$. The value $s = 2$ gives $a = 0$. Both families satisfy $R_a(s^2 - 4) = 0$ by direct substitution, so the list is complete. The member $a = -16$ is $s = 0$ in the first family; every other member is irreducible by (i).

It remains to separate C_4 from D_4. Both groups occur with non-square discriminant, and both have a resolvent with exactly one rational root. Let r be that root and let $L = \mathbb{Q}(u)$ with $u^2 = r$. Ferrari's identity factors $f_{4,a}$ over L as a product of two quadratics $(x^2 + ux + v)(x^2 - ux + w)$, and the splitting field is $L\big(\sqrt{\Delta_1}\big)$ where $\Delta_1 = -r - 2a/u$ is the discriminant of the first factor.

This extension of $\mathbb{Q}$ is Galois of degree 4 if and only if it is stable under $u \mapsto -u$, if and only if $\sigma(\Delta_1)/\Delta_1$ is a square in $L^\times$. A direct computation gives

$$\frac{\sigma(\Delta_1)}{\Delta_1} = \frac{ru - 2a}{ru + 2a}.$$

The Galois group then has order 4. It is not V_4, because the resolvent does not split completely, so it is C_4. Otherwise the splitting field has degree 8 and the group is D_4.

Write $m = s - 2$ and $r = m(m + 4)$, and take first $a = -m^2(m + 4)$. The ratio becomes $(u + 2m)/(u - 2m)$. Setting this equal to $(\alpha + \beta u)^2$ with $\alpha, \beta \in \mathbb{Q}$ and equating coefficients of 1 and u produces the system

$$\begin{aligned} \alpha^2 + r\beta^2 - 4m\alpha\beta &= 1, \\ \alpha\beta &= -\frac{2}{3m - 4} \qquad (m \neq 4/3). \end{aligned}$$

Eliminating β yields a quadratic equation for $Z = \alpha^2$ whose discriminant is a square, with roots

$$Z = -\frac{m + 4}{3m - 4}, \qquad Z = -\frac{4m}{3m - 4}.$$

For Z to be a square in $\mathbb{Q}$ one of these two rational numbers must be a square. In the second case, $-4m/(3m - 4) = p^2/q^2$ in lowest terms rearranges to $m = 4p^2/(3p^2 + 4q^2)$, whose denominator is positive, so $0 \leq m \leq 4/3$. The only positive integer value is $m = 1$, attained at $(p, q) = (2, 1)$, and it gives $Z = 4 = 2^2$. The corresponding parameter is $s = 3$ and $a = -5$. In the first case, $-(m + 4)/(3m - 4) = p^2/q^2$ rearranges to $m = 4(p^2 - q^2)/(3p^2 + q^2)$, which lies in the interval $(-4, 4/3]$. Checking the integers $m \in \{-3, -2, -1, 1\}$ shows that none of them makes the ratio a square, and $m = -4$ forces $\alpha = 0$, which fails the linear equation. Thus, in this family, the ratio is a square only for $a = -5$.

For the second family $a = m(m + 4)^2$ with $m > 0$, the same computation produces

$$\alpha^2 = -\frac{m}{3m + 16} \qquad \text{or} \qquad \alpha^2 = -\frac{4(m + 4)}{3m + 16},$$

both negative for $m > 0$, so neither is a square in $\mathbb{Q}$. The group is therefore D_4 for every irreducible member except $a = -5$, where it is C_4. □

Theorem 6. *If $f_{5,a}$ is irreducible, then* $\mathrm{Gal}(f_{5,a}) \cong S_5$.

The proof uses Dummit's resolvent [2]. For $x^5 + Ax + B$ it is the sextic

$$\begin{aligned} &t^6 + 8At^5 + 40A^2t^4 + 160A^3t^3 + 400A^4t^2 \\ &\quad + (512A^5 - 3125B^4)t + (256A^6 - 9375AB^4). \end{aligned} \tag{1}$$

Dummit's theorem says that an irreducible quintic $x^5 + Ax + B \in \mathbb{Q}[x]$ is solvable by radicals if and only if (1) has a rational root, in which case the Galois group is C_5, D_5, or the Frobenius group F_{20} of order 20. We checked the criterion against a direct Galois-group computation for all irreducible $x^5 + Ax + B$ with $0 < |A|, |B| \leq 60$: 14130 polynomials, 126 of them solvable, and no disagreement.

Lemma 7. *For $a \neq 0$,* $\mathrm{disc}(f_{5,a}) = a^4(3125 - 256a)$ *is not a square.*

Proof. The exponent of a is even, so the discriminant is a square if and only if $3125 - 256a = m^2$ for some integer m. Reducing modulo 256 gives $m^2 \equiv 3125 \equiv 53 \pmod{256}$. The squares modulo 256 were compared with 53 for every residue class; none agrees. □

Proof of Theorem 6. By Lemma 7 the discriminant is not a square, so the Galois group is not contained in A_5. The transitive subgroups of S_5 not contained in A_5 are S_5, the dihedral group of order 10, and F_{20}; the last two are solvable. It is therefore enough to show that (1), evaluated at $A = B = -a$, has no rational root.

Substituting $A = B = -a$ and $t = az$ factors the sextic as $a^5\big(a\,P(z) + 3125(3 - z)\big)$, where

$$P(z) = (z - 2)^4(z^2 + 16).$$

A root $z = 3$ gives a arbitrary in the factor a^5 but $t = 3a$ is a root of the sextic only for $a = 0$. For $z \neq 2$ one obtains $a = 3125(z - 3)/P(z)$. Write $z = u/v$ in lowest terms with $v > 0$, and set $d = u - 2v$. Then $\gcd(d, v) = 1$ and

$$a = \frac{3125\,(d - v)\,v^5}{d^4\,(d^2 + 4dv + 20v^2)}.$$

The denominator is coprime to v: indeed $\gcd(d, v) = 1$ and $\gcd(d^2 + 4dv + 20v^2, v) = \gcd(d^2, v) = 1$. Hence $d^4(d^2 + 4dv + 20v^2)$ divides $5^5(d - v)$. Every prime divisor of d other than 5 would then divide $d - v$, hence divide v, which is impossible. So $d = \pm 5^\alpha$.

Write $N = d^2 + 4dv + 20v^2$. The same coprimality gives $|d|^4 N \leq 5^5|d - v|$. Since $N = (d + 2v)^2 + 16v^2 \geq 16v^2$,

$$16\,|d|^4\,v^2 \leq 3125\big(|d| + v\big).$$

If $|d| \geq 25$, already $v = 1$ violates the inequality, and larger v makes it worse. If $|d| = 5$, the inequality forces $v = 1$. If $|d| = 1$, it forces $v \leq 200$. Testing these finitely many pairs (d, v) leaves only $(d, v) = (1, 1)$, which gives $a = 0$. Thus no nonzero integer a makes the sextic solvable, and the Galois group is S_5. □

For completeness, two reducible members are $f_{5,-1} = (x^2 + x + 1)(x^3 - x^2 + 1)$ and $f_{5,32} = (x + 2)(x^4 - 2x^3 + 4x^2 - 8x - 16)$. We do not claim that the list is exhaustive.

5 The unit $\theta + 1$

Proposition 8. *If $f_{n,a}$ is irreducible and θ is a root, then $\theta + 1$ is a unit of $K_{n,a}$ of norm 1.*

Proof. For a monic minimal polynomial, $\mathrm{N}(\theta - \lambda) = (-1)^n f_{n,a}(\lambda)$. Hence

$$\mathrm{N}(\theta + 1) = (-1)^n f_{n,a}(-1) = (-1)^n\big((-1)^n + a - a\big) = 1.$$

A unit of norm ± 1 lying in $\mathcal{O}_K$ is a unit, and the norm is $+1$. □

Primitivity is a different question: $\theta + 1$ may be a proper power of a fundamental unit. The phenomenon is rare for squarefree a, and in degree 3 it does not occur in a long initial range.

Theorem 9. *Let $n = 3$ and let a be squarefree with $2 \leq |a| \leq 33535$. Then $\theta + 1$ is not a proper power in $\mathcal{O}_K^\times$. Equivalently, its image is nonzero in $\mathcal{O}_K^\times/\{\pm 1\}$ modulo every proper subgroup of finite index.*

Proof. There are 40772 such fields. Let $\eta = \theta + 1$ and let d_K be the field discriminant. For 40615 of them $|d_K|$ exceeds $2^6\bigl(1 + (2|a|)^{1/2}\bigr)^3$. In a complex cubic field the conjugates of a unit ε satisfy $\prod |\sigma\varepsilon| = 1$, and the discriminant of the order $\mathbb{Z}[\varepsilon]$ is an integer multiple of d_K. Comparing $\prod_{i<j} |\varepsilon_i - \varepsilon_j|$ with the house of η shows that a representation $\eta = \pm\varepsilon^k$ with $k \geq 2$ would force $|d_K|$ to lie under that bound. This is the standard house estimate coming from $|\varepsilon_i - \varepsilon_j| \leq 2 \max |\varepsilon_\ell|$ and $\max |\varepsilon_\ell| \leq \bigl(1 + |\theta|_{\max}\bigr)^{1/k}$, using $|\theta|^2 \leq 2|a|$ at the real place; we omit the arithmetic, which is finite and was checked exactly.

The remaining 157 fields were handled directly. For 156 of them the exponent k permitted by the same estimate is at most an explicit bound $k_{\max} \leq 30$, and for every prime $p \leq k_{\max}$ the polynomials $x^p - \eta$ and $x^p + \eta$ have no root in K. Roots were computed with `nfroots`, which does not depend on the generalized Riemann hypothesis. The last field, $a = -2$, has discriminant small enough that `bnfcertify` certifies the full unit group unconditionally; the exponent of η on that basis is 1. □

Outside this range, and in higher degree, we only report computations conditional on GRH. For $2 \leq n \leq 7$ and $|a|$ up to several hundred, $\theta + 1$ fails to be primitive only for thin sets of a, typically perfect powers; the cubic exceptions with $|a| \leq 3000$ are listed in the file `computations/check_units.gp`.

6 Genus theory

Throughout this section, a is squarefree, $\gcd(a, n) = 1$, and $f_{n,a}$ is irreducible. For each prime $q \mid a$ write $d_q = \gcd(q - 1, n)$. If $d_q > 1$, let k_q be the unique subfield of $\mathbb{Q}(\zeta_q)$ of degree d_q, and let F_a be the compositum of these fields k_q. Set $g^+(a) = \prod_{d_q > 1} d_q$, and $g^+(a) = 1$ if no such q exists. The field F_a is totally real if and only if $(q - 1)/d_q$ is even for every q under the product, since k_q is the unique subfield of degree d_q and it is real precisely then.

Theorem 10. *Assume in addition that* $\mathrm{Gal}(f_{n,a}) \cong S_n$ *with* $n \geq 3$. *Then* KF_a/K *is unramified at every finite place,* $K \cap F_a = \mathbb{Q}$, *and*

$$[KF_a : K] = g^+(a).$$

This degree divides the narrow class number of K. *It divides the ordinary class number if* F_a *is totally real or if* K *has no real place. Otherwise* $g^+(a)$ *is even and* $g^+(a)/2$ *divides the ordinary class number.*

Proof. By Theorem 2(i), every $q \mid a$ is totally ramified in K: here $k = v_q(a) = 1$, so $g = 1$ and $q\mathcal{O}_K = \mathfrak{q}^n$. The extension $k_q/\mathbb{Q}$ is cyclic of degree d_q, ramified only at q, and tamely, because $d_q \mid (q - 1)$. In the compositum Kk_q the ramification index of q over $\mathbb{Q}$ is $\mathrm{lcm}(n, d_q)$. The hypothesis $\gcd(a, n) = 1$ keeps q from dividing n, and $d_q \mid n$, so the least common multiple equals n. The relative ramification index over K is therefore 1. No other finite prime ramifies in k_q, so Kk_q/K is unramified at every finite place. The same holds for the compositum F_a.

The field K has no proper subfield. Indeed the point stabilizer $S_{n-1} \subset S_n$ is a maximal subgroup, so the stem field of an S_n-extension of degree n has no intermediate field. An abelian extension of $\mathbb{Q}$ cannot contain K, because $K/\mathbb{Q}$ is not Galois. Hence $K \cap F_a = \mathbb{Q}$ and $[KF_a : K] = [F_a : \mathbb{Q}]$. The fields k_q are linearly disjoint because they ramify at distinct primes, so $[F_a : \mathbb{Q}] = \prod d_q$.

An abelian extension of K unramified at the finite places is contained in the narrow class field, and its degree divides the narrow class number. If F_a is totally real, every real place of K stays

real in KF_a, so there is no ramification at infinity either, and the degree divides the ordinary class number. If K has no real place, there is nothing at infinity to ramify. If some k_q is imaginary, then d_q is even: otherwise d_q would divide $(q-1)/2$. Thus $g^+(a)$ is even. Complex conjugation acts on F_a through a quotient of order 2, and the fixed field of a lift of conjugation cuts KF_a/K down to a subextension of index 2 which is unramified at the infinite places as well. □

For $n = 3, 4, 5$, Theorems 4–6 supply the hypothesis $\mathrm{Gal} \cong S_n$ for every irreducible member outside an explicit list, so the divisibility is unconditional for those fields. In the cyclic cubic case the same construction overcounts by a factor of 3, because K itself sits inside the genus field; Ishida's formula [3] gives divisibility by 3^{t-1} rather than 3^t, where t is the number of primes $q \mid a$ with $q \equiv 1 \pmod 3$.

The genus number divides the class number, but the classes it produces need not be the classes of the ramified primes. For $a = 31$ and $n = 3$, one has $g^+ = 3$ and the class group is $(\mathbb{Z}/3\mathbb{Z})$, while the unique ramified prime is principal. A separate local condition controls the subgroup generated by those primes.

Proposition 11. *Keep the hypotheses of this section, and write $\mathfrak{q}$ for the unique prime of K above $q \mid a$. Suppose $\prod_{q|a} \mathfrak{q}^{c_q} = (\beta)$ for some $\beta \in K^\times$ and integers c_q. Then for every $q \mid a$ with $d_q > 1$, the class of*

$$(-1)^{c_q(n+1)} \prod_{p|a,\, p\neq q} p^{c_p - c_q}$$

in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{d_q}$ lies in the image of $\mathcal{O}_K^\times \to \mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{d_q}$.

Proof. From $\theta^n = a(\theta+1)$ and $v_{\mathfrak{q}}(\theta) = 1$ one gets $(\theta) = \prod_{p|a} \mathfrak{p}$. The ideal equation $\prod \mathfrak{q}^{c_q} = (\beta)$ implies $\beta^n = u \prod_{p|a} p^{c_p}$ for a unit u, because both sides generate $\prod \mathfrak{q}^{nc_q} = \left(\prod p^{c_p}\right)$. Set $\delta = \beta\, \theta^{-c_q}$. Then $v_{\mathfrak{q}}(\delta) = 0$, and substituting $\theta^n = a(\theta+1)$ gives

$$\delta^n\, (\theta+1)^{c_q} = u\, (-1)^{c_q(n+1)} \prod_{p\neq q} p^{c_p - c_q}$$

up to the sign absorbed in $a = \pm \prod p$. Reducing modulo $\mathfrak{q}$ kills θ, so $\theta + 1 \equiv 1$, and $\overline{\delta}^n$ is an nth power in $\mathbb{F}_q^\times$. The group of nth powers has index d_q, and $\overline{u}$ is the image of a global unit. □

The proposition is the exact obstruction used to build a Rédei matrix. Let $q_1, \ldots, q_t$ be the prime divisors of a with $d_{q_i} > 1$, and let $p_1, \ldots, p_s$ be all the prime divisors of a. Choosing a primitive root modulo q_i, one obtains a matrix $A \in \prod_i \mathbb{Z}/d_{q_i}\mathbb{Z}$ whose columns are the discrete logarithms of the p_j, with the diagonal entry replaced by the logarithm of $(-1)^{n+1}a/q_i$ so that Proposition 11 becomes the statement Ac lies in the image of the unit group. Ignoring free units and retaining only the torsion unit -1 produces a concrete lower bound. That bound is not a formal consequence of the proposition, because free units may enlarge the relation lattice. It is, however, what the class group computations confirm.

7 The cyclic members and simplest cubics

Theorem 12. *Let $t \in \mathbb{Z}$ and let $a = t^2 + 3t + 9$. Let ρ be a root of Shanks' polynomial $g_t = x^3 - tx^2 - (t+3)x - 1$, and let $\rho' = -1/(1+\rho)$ be its Galois conjugate. Then $\rho' - \rho$ is a root of $f_{3,a}$, and $K_{3,a} = \mathbb{Q}(\rho)$ is the simplest cubic field of parameter t.*

Proof. The polynomial g_t is irreducible: the only possible rational roots are ± 1, and $g_t(1) = -2t-3$, $g_t(-1) = 1$. Its discriminant is $(t^2+3t+9)^2$, so the extension is cyclic, and the standard identity $\rho \mapsto -1/(1+\rho)$ generates the Galois group [10]. Set $\delta = -x - 1/(1+x)$ in $\mathbb{Q}(x)/(g_t)$. Reducing $\delta^3 - a\delta - a$ via g_t gives zero, by a direct expansion of the numerator $(1+x)^3\big(\delta^3 - a\delta - a\big)$ and cancellation against g_t. Thus δ is a root of $f_{3,a}$. Theorem 4 says that $f_{3,a}$ is irreducible of discriminant a square, so $\mathbb{Q}(\delta)$ is the cyclic cubic field of discriminant dividing $a^2(2t+3)^2$, which is the field $\mathbb{Q}(\rho)$. □

For these fields $\theta + 1$ is not merely a unit: it is the square of a Shanks unit, up to the Galois action. With ρ' as above one has $\theta + 1 = \rho^2\rho'$, and $\mathrm{N}(\theta+1) = 1$ agrees with Proposition 8. The index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$ is generally greater than one. For $t = 1$, $a = 13$, one has $\mathrm{disc}(f) = 5^2 \cdot 13^2$ and $d_K = 13^2$, so the index is 5.

8 Computations

All computations were done in PARI/GP 2.15.4 [9]. Class groups and unit groups computed with `bnfinit` are conditional on GRH; the local statements, the Galois groups in degrees ≤ 7, the identity of Theorem 12, and Theorem 9 do not use GRH. `bnfcertify` was used for the single cubic field not reached by the house bound or by `nfroots`.

The splitting types and index valuations of Theorem 2 and Theorem 3 were compared with `idealprimedec` and `nfdisc` for every irreducible $f_{n,a}$ with $2 \leq n \leq 9$ and $1 \leq |a| \leq 400$. This is 6388 fields and 25287 pairs (p, K) in which p divides $\mathrm{disc}(f_{n,a})$. The 591 pairs in which a prime divides $\gcd(v_p(a), n)$ were excluded, as the theorems require. No mismatch occurred.

Galois groups were computed with `polgalois`. In degree 4, every a with $|a| \leq 20000$ falls under Theorem 5, and the only non-symmetric group is D_4 or the single field $a = -5$. In degree 5, every irreducible $f_{5,a}$ with $|a| \leq 20000$ has group S_5, in agreement with Theorem 6. In degree 6, with $|a| \leq 3000$, the group is S_6 except for the six parameters

a	$\lvert\mathrm{Gal}\rvert$	transitive label
$-320, -125, -3$	72	$S_3 \wr C_2$
-116	120	$\mathrm{PGL}_2(\mathbb{F}_5)$
49	24	$C_2 \wr C_3$
2662	48	$C_2 \wr S_3$.

In degree 7, every irreducible $f_{7,a}$ with $|a| \leq 1500$ has group S_7.

Table 1 illustrates Theorem 10 in degree 3. Here t is the number of primes $q \mid a$ with $q \equiv 1 \pmod 3$, the genus number is 3^t in the S_3 case and 3^{t-1} in the C_3 case, and the column "Rédei" is the order of the subgroup generated by the ramified primes, as predicted by the matrix of Proposition 11 with only the torsion unit retained. The class groups themselves are conditional on GRH. The last column records whether $\theta + 1$ is primitive.

The entry $a = 31$ is the example in which the genus contribution is real but the ramified prime is principal. The entry $a = 55$ has trivial genus number and a class group of order 3 coming from elsewhere. The entry $a = 1729 = 7 \cdot 13 \cdot 19$ is a cyclic field whose class group is exactly the genus group $(\mathbb{Z}/3\mathbb{Z})^2$.

For squarefree a coprime to n, with $n = 3$ and $|a| \leq 1500$, $n = 4$ and $|a| \leq 400$, $n = 5$ and $|a| \leq 150$, and $n = 6$ and $|a| \leq 60$, the narrow class number was divisible by $g^+(a)$ and the ordinary class number by the number in Theorem 10, in all 1878 fields for which the Galois group was S_n.

Table 1: Cubic fields $K_{3,a}$. Class groups are conditional on GRH.

a	d_K	group	Cl	t	genus	Rédei	h
7	49	C_3	1	1	1	1	1
13	169	C_3	1	1	1	1	1
217	47089	C_3	[3]	2	3	3	3
1729	2989441	C_3	[3, 3]	3	9	9	9
31	93217	S_3	[3]	1	3	1	3
14	5684	S_3	[3]	1	3	3	3
91	2790697	S_3	[6, 3]	2	9	3	18
−91	−3237871	S_3	[15, 3]	2	9	3	45
10	1300	S_3	1	0	1	1	1
55	583825	S_3	[3]	0	1	1	3

The matrix bound of Proposition 11 was a lower bound for the subgroup generated by the ramified primes in every one of those fields. Both checks use `bnfinit` and are conditional on GRH.

A larger cubic sample, still conditional on GRH, counts the 3-rank against the genus number. Among the 4511 fields with $n = 3$, squarefree a, $3 \nmid a$, and $2 \le |a| \le 5000$ whose Galois group is S_3, the 3-rank r and the number t of admissible primes satisfy $r \ge t$ in every case, as Theorem 10 requires. The distribution is recorded in Table 2. The Rédei bound was nontrivial for 1774 of these fields and equal to the order of the subgroup generated by the ramified primes for 1359 of them.

Table 2: 3-ranks of S_3 fields $K_{3,a}$ with squarefree a, $3 \nmid a$, and $|a| \le 5000$. Conditional on GRH.

t	fields	$r = t$	$r = t + 1$
0	1724	807	733
1	2238	1569	613
2	537	426	109
3	12	11	1

The twelve fields with $t = 3$ are the interesting ones: eleven have 3-rank exactly 3, so the genus group accounts for the whole 3-primary part, and one has 3-rank 4.

9 Remarks

The excluded local case $p \mid \gcd\big(v_p(a), n\big)$ is the natural place for a sequel. The Newton polygon is still a single side, but the residual polynomial $Y^g - u$ is inseparable, and the index depends on the residual factorization of a refinement. A uniform formula in that case would complete Theorem 3.

A second question is whether $\theta + 1$ generates a finite-index subgroup of the unit group whose index stays bounded for squarefree a. Theorem 9 says that the index is 1 throughout a long cubic range. The GRH computations suggest the same pattern in degrees 4 and 5 once perfect powers are removed, but the house argument loses a power of $|a|$ as soon as there are two real places, and we do not have a proof.

References

[1] B. Bensebaa, A. Movahhedi, and A. Salinier, The Galois group of $X^p + aX^s + a$, Acta Arith. **134** (2008), 55–65.

[2] D. S. Dummit, Solving solvable quintics, Math. Comp. **57** (1991), 387–401.

[3] M. Ishida, On the genus field of an algebraic number field of odd prime degree, J. Math. Soc. Japan **27** (1975), 289–293.

[4] M. Ishida, *The genus fields of algebraic number fields*, Lecture Notes in Math. **555**, Springer, 1976.

[5] R. Ibarra, H. Lembeck, M. Ozaslan, H. Smith, and K. E. Stange, Monogenic fields arising from trinomials, preprint.

[6] L. Jones and D. White, Monogenic trinomials with non-squarefree discriminant, Internat. J. Math. **32** (2021), no. 13, 2150089.

[7] K. J. McGown, F. Thorne, and A. Tucker, Counting quintic fields with genus number one, preprint, `arXiv:2006.12991`.

[8] Ö. Ore, Newtonsche Polygone in der Theorie der algebraischen Körper, Math. Ann. **99** (1928), 84–117.

[9] The PARI Group, PARI/GP version 2.15.4, Univ. Bordeaux, 2022, `http://pari.math.u-bordeaux.fr/`.

[10] D. Shanks, The simplest cubic fields, Math. Comp. **28** (1974), 1137–1152.

[11] B. K. Spearman and K. S. Williams, Characterization of solvable quintics $x^5 + ax + b$, Amer. Math. Monthly **101** (1994), 986–992.